

# ChromaTau: Conditioning State-Space Timescales on Three-Dimensional Chromatin Structure

Author One and Author Two

Affiliation, {a.one, a.two}@institution.edu

## Abstract

We introduce **ChromaTau**, a genomic sequence model that conditions the retention timescale of a selective state-space model on measured three-dimensional chromatin structure during pretraining. Existing structure-aware genomic models attach Hi-C to representations that are already frozen, which cannot change what the encoder learns. ChromaTau instead injects structure at the one place in a selective SSM where long-range behaviour is decided — the timescale  $\Delta$ , which sets how long the recurrent state retains information — so that a model told where domain boundaries lie can learn to reset at them and retain within them. Structure is used only during pretraining and removed at transfer, so any benefit resides in the learned sequence weights.

Two design problems had to be solved to make the mechanism expressible, and both are contributions in their own right. First, we identify a *timescale ceiling* present before training begins: retention obeys  $\tau = 1/(\Delta|A| + p)$ , so initialisation constants impose a hard upper bound on memory. Mamba’s reference  $\Delta_{\min} = 10^{-3}$  caps  $\tau$  at  $10^3$  tokens — at single-nucleotide tokenisation, shorter than a typical gene; we measure 999.5. We remove it at *zero* parameter cost, moving median trained  $\tau$  from 14.2 to 434.7 and taking layer-directions reaching domain scale from 0/32 to 32/32. We further show the ceiling recurs through a second constant that the standard  $\tau$  diagnostic omits entirely.

Second, we show that a *per-position* structural bias is the wrong instrument, and quantify why: only 11.0% of the structural signal’s variance lies within a 65,536 bp window. We establish this is intrinsic to Hi-C spatial autocorrelation rather than an artefact of binning or context length, by measuring it across window widths from 8 kb to 524 kb and bin resolutions from 5 kb to 1 kb. ChromaTau therefore conditions at two scales jointly, local and window-pooled, for +0.43% parameters — the configuration with access to the full structural signal rather than a tenth of it.

## 1 Introduction

The genome is read as a one-dimensional string but functions as a folded three-dimensional object. Regulatory contacts routinely join loci hundreds of kilobases apart in sequence that are adjacent in space, and this folding is measurable: chromosome conformation capture (Hi-C) yields contact maps from which topologically associating domains (TADs) and A/B compartments can be derived [4, 6].

Sequence models for genomics have grown contexts aggressively, first with efficient attention and more recently with state-space models (SSMs) whose cost is linear in length [2, 5, 7]. A natural question follows: if a model must learn long-range dependence, and that dependence is partly explained by measured folding, does supplying the folding help?

Existing work answers this by *post-hoc attachment*. Structure is applied to representations that are already frozen, typically through graph attention over a contact graph. This tests whether structure is useful for a downstream readout, but it cannot test whether structure changes what the encoder learns.

We ask the other question. We inject structure *during* pretraining, at the one place in a selective SSM where long-range behaviour is actually decided: the timescale  $\Delta$ , which sets how long the recurrent state retains information. The hypothesis is direct — a model told where domain boundaries are should learn to reset at them and retain within them.

Building it required solving two problems that we believe are of independent interest, and the second of them determines the architecture. Our contributions are:

- **ChromaTau**, a genomic sequence model that conditions SSM retention timescales on measured chromatin structure *during* pretraining (§6). Structure is supplied only while learning and removed at transfer, so any benefit must reside in the sequence weights. The pathway costs +0.43% parameters and is zero-initialised, making the model numerically identical to a sequence-only baseline at step 0.
- **A measurement that determines the design** (§4). Only 11.0% of the structural signal’s variance lies within a 65,536 bp window. We show across seven window widths and three bin resolutions that this is intrinsic to Hi-C spatial autocorrelation, not an artefact — so a per-position coupling can reach only a tenth of the signal by construction. ChromaTau conditions at two scales jointly as a direct consequence.
- **Removal of a timescale ceiling** that makes the mechanism expressible at all (§5). The reference  $\Delta$  initialisation caps retention at  $1/\Delta_{\min}$  tokens before the first gradient step, at zero parameter cost to fix. We show the ceiling recurs through a second constant that the standard  $\tau$  diagnostic omits, and that a model can therefore report a plausible memory horizon that is not its own.
- **A pre-registered evaluation protocol** (§7) with matched parameters, matched compute, paired initialisation, cross-chromosome splits, and input-substitution controls that separate *reliance* on structure from *benefit* from it.

Every threshold in this paper was fixed before the corresponding number existed, and none was moved afterwards.

## 2 Related work

**Long-context genomic sequence models.** Attention-based genomic models are limited by quadratic cost, which at single-nucleotide tokenisation binds quickly. Implicit-convolution and state-space architectures relaxed this: HyenaDNA [5] reaches  $10^6$ -token contexts at single-nucleotide resolution, and Caduceus [7] adds reverse-complement equivariance to a bidirectional Mamba backbone [2]. Our backbone is deliberately of this family and parameter-matched to the smallest published Caduceus configuration, so that the mechanism rather than the scale is what differs. The timescale ceiling of §5 is a property of the shared initialisation and therefore applies to this family generally, not only to the model studied here.

**Structure-aware genomic models.** A second line predicts folding *from* sequence: Akita and Orca learn contact maps directly, establishing that sequence carries substantial information about three-dimensional organisation. The converse direction — using measured folding to improve sequence representations — has been approached by attaching structure to representations that are already trained. Graph-attention methods of this kind operate over a contact graph on top of *frozen* embeddings. This is an effective way to exploit structure at readout time, but it leaves the encoder unchanged and therefore cannot answer whether structure alters what is learned. Our design differs in exactly this respect: conditioning is applied during pretraining and is removed at transfer time, so any benefit must reside in the learned sequence weights.

**Conditioning recurrent timescales.** Gating a recurrent model’s retention on side information has precedent, where forget-gate biases control effective memory length. What is specific here is interpretability: in a selective SSM the timescale is an explicit quantity in tokens, so a signal predicting *where dependencies end* — which is what an insulation profile is — can be applied to it directly and audited afterwards.

### 3 Background

**Selective state-space models.** A selective SSM maintains a state  $h_t$  updated as  $h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$ , where the discretisation  $\bar{A}_t = \exp(\Delta_t A)$  makes the effective retention time input dependent. With  $A$  diagonal and negative, the characteristic decay time of direction  $i$  is

$$\tau_{t,i} = \frac{1}{\Delta_t |A_i|}, \quad (1)$$

so  $\Delta$  is the quantity that decides how far back the model can see. In Mamba,  $\Delta_t = \text{softplus}(W_\Delta x_t + b_\Delta)$ , and  $b_\Delta$  is initialised so that  $\Delta$  spans  $[\Delta_{\min}, \Delta_{\max}]$ .

**Structural features.** From a Hi-C contact matrix at 5 kb resolution we derive an eight-dimensional per-bin vector  $\phi$ : insulation at three scales (100, 250, 500 kb), a 2 Mb directionality index, log contact density, upstream mass fraction, short-to-long range ratio, and the first principal component of the correlation matrix (the A/B compartment eigenvector). Validated against independently processed tracks, our insulation attains Pearson  $r = 0.9969$  and our compartment eigenvector  $r = 0.9759$ .

Under reverse complementation the features carry signs  $[1, 1, 1, -1, 1, -1, 1, 1]$ ; directionality and upstream mass fraction are antisymmetric. This flip must be applied to raw  $\phi$  before encoding, since an encoder mixes the coordinates and destroys the correspondence.

### 4 Where the structural signal lives, and what it implies for design

A structural signal can only inform a per-position bias to the extent that it *varies* across positions the model sees together. Before designing the conditioning pathway we therefore measured how much of  $\phi$ ’s variance is actually available inside a window, defining the reachable fraction as

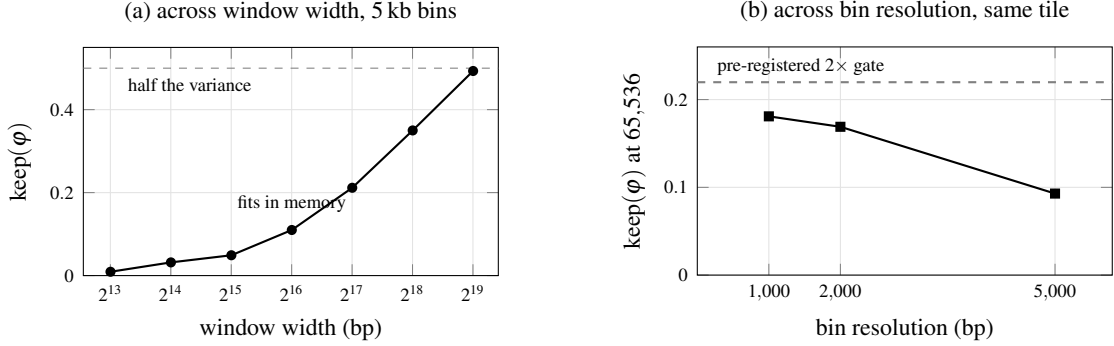
$$\text{keep}(\phi) = \frac{\mathbb{E}_w [\text{Var}_{t \in w}(\phi_t)]}{\text{Var}_t(\phi_t)}. \quad (2)$$

On held-out windows  $\text{keep}(\phi) = 0.0573$  at 32,768 bp and 0.1099 at 65,536 bp: roughly 90% of the structural variance lies *between* windows. A per-position bias is therefore driven by an input that is nearly constant over the interval it modulates — a ceiling imposed by the window size itself, not by training.

**This is the measurement that determines the architecture of §6.**

Two natural remedies fail, and we measured both rather than assuming (Fig. 1).

**Widening the window is a partial mitigation, not a fix.** Across widths from 8 kb to 524 kb,  $\text{keep}(\phi)$  rises monotonically but slowly. At 65,536 bp — which we verify fits in 7.51 GiB with gradient checkpointing — 89.0% of the variance is still between windows. At 131,072 bp, which also fits (14.90 GiB) at  $2.41 \times$  the step cost, 78.8% remains. Even at 524,288 bp, beyond practical memory, half of it does.



**Figure 1:** The signal is between windows at every reachable setting. (a) Widening the context raises  $\text{keep}(\phi)$  but even at 524,288 bp — far beyond what fits in memory — half the variance is still between windows. (b) Refining bins five-fold on a fixed tile raises it by  $1.95\times$ , missing a gate fixed in advance at  $2\times$ ; at 1 kb, 81.9% of the variance remains between windows.

**Refining the bins does not fix it either.** We fetched a 10 Mb tile at 1 kb and 2 kb and recomputed  $\phi$  with the same pipeline, against a same-tile 5 kb control. Like-for-like, 5 kb  $\rightarrow$  1 kb raises  $\text{keep}(\phi)$  at 65,536 bp by  $1.95\times$  ( $0.0929 \rightarrow 0.1809$ ), missing a gate fixed in advance at  $2\times$ . Most of the gain arrives by 2 kb: the final step to 1 kb buys  $1.071\times$  for  $1.70\times$  the contact entries. At 1 kb bins — five times finer than any resolution used here — 81.9% of the variance is *still* between windows.

**The encoder is not the bottleneck.** One might suspect the two-dimensional structural code discards within-window variation. It does the opposite: the ratio of keep after encoding to keep of raw  $\phi$  is  $2.21 / 2.35 / 1.20$  across seeds, mean 1.92. The encoder *increases* the within-window variance fraction of what passes through it.

**Design consequence.** Three independent interventions — context length, bin resolution and the encoder — leave the conclusion unchanged: the dominant variation in Hi-C-derived features occurs on scales longer than any context these models can process. A purely per-position coupling therefore addresses the minority share by construction, while a purely per-window coupling has the opposite defect, carrying the between-window share but discarding local variation. **ChromaTau conditions at both scales jointly** (§6), the only configuration with access to all of  $\phi$  rather than a fraction fixed in advance by the window size.

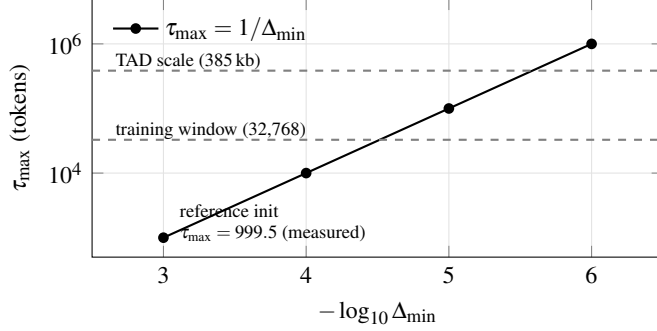
## 5 A timescale ceiling before the first gradient step

Equation (1) has an immediate consequence that we have not seen stated. If  $A_i = -(i+1)$  for  $i = 0, \dots, d_{\text{state}} - 1$ , as in the reference initialisation, then  $|A_i| \geq 1$  and therefore

$$\tau_{\max} = \max_{t,i} \frac{1}{\Delta_t |A_i|} \leq \frac{1}{\Delta_{\min}}. \quad (3)$$

The bound involves no learned quantity. It is a property of the initialisation, present before any data is seen, and no amount of training can exceed it while  $\Delta$  remains bounded below.

With Mamba’s reference  $\Delta_{\min} = 10^{-3}$  this is 1,000 tokens. At one token per base pair — the tokenisation used by recent genomic SSMs — that is shorter than a typical human gene and two orders of magnitude below a TAD. We measure  $\tau_{\max} = 999.5$  at initialisation, matching Eq. (3).



**Figure 2:** The ceiling of Eq. (3). Retention time cannot exceed  $1/\Delta_{\min}$  regardless of training. At the reference  $\Delta_{\min} = 10^{-3}$  the bound sits two orders of magnitude below the training window and three below domain scale.

**Table 1:** Effect of the initialisation fix. Three seeds per row, 2,000 steps, 32,768 bp window. Parameter count is identical across rows.

|                                         | median trained $\tau$ | directions reaching TAD scale | val bits/nt         |
|-----------------------------------------|-----------------------|-------------------------------|---------------------|
| reference ( $\Delta_{\min} = 10^{-3}$ ) | 14.2                  | 0/32                          | $1.5210 \pm 0.0040$ |
| corrected ( $\Delta_{\min} = 10^{-6}$ ) | <b>434.7</b>          | <b>32/32</b>                  | $1.5197 \pm 0.0025$ |

**The fix, and what it costs.** Lowering  $\Delta_{\min}$  to  $10^{-6}$  raises the bound to  $10^6$  tokens. This changes three constants and *no parameters*. Table 1 reports trained models before and after.

**A second constant, and why the diagnostic missed it.** Our conditioned model (§6) adds a permeability gate  $p_t$ , giving  $\tau = 1/(\Delta|A| + p)$ . A floor on  $p$  is a ceiling on  $\tau$  exactly as  $\Delta_{\min}$  is, and with the gate bias initialised at  $-4$  we have  $p = 0.0182$  and hence  $\tau \leq 1/p \approx 55$  tokens — *below* the uncorrected reference bound, and applying only to the conditioned arm.

This went undetected because the standard estimator computes  $\tau$  from  $\Delta$  alone, omitting  $p$ . Under that estimator the conditioned model reported a median  $\tau$  of 487.3 against the baseline’s 434.7, i.e. an apparent *lengthening*; recomputed with  $p$  included, its true median is  $\approx 51$  tokens, roughly  $8.5\times$  *shorter*. The quantity being reported was not the quantity governing memory.

We state this as a general caution. In any selective SSM, every additive term entering the discretisation imposes its own reciprocal ceiling, and a  $\tau$  diagnostic that omits one will report a plausible number that is not the model’s memory. We found two such constants in one architecture.

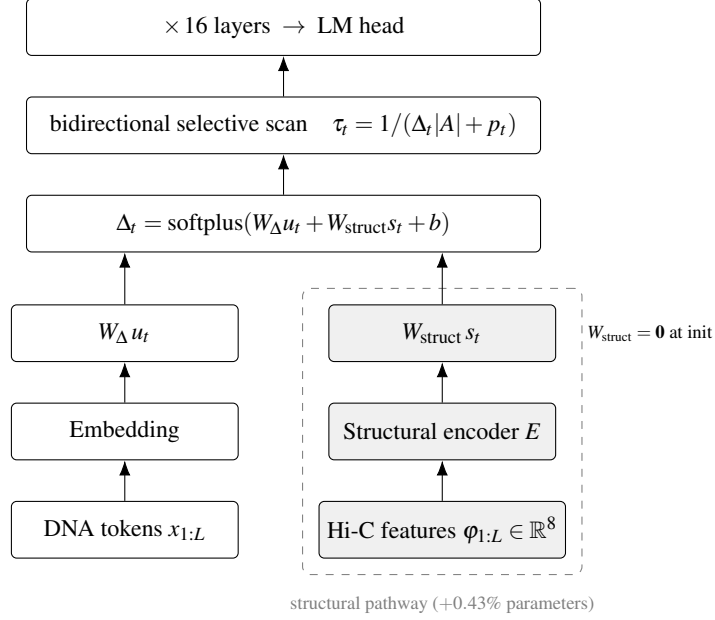
## 6 The ChromaTau model

The backbone is a bidirectional Mamba masked language model:  $d_{\text{model}} = 256$ , 16 layers,  $d_{\text{state}} = 16$ , expansion 2, one token per base pair, trained to reconstruct masked nucleotides. The uniform-random loss floor is exactly 2.000 bits/nt.

Structural conditioning adds a pointwise encoder  $E : \mathbb{R}^8 \rightarrow \mathbb{R}^2$  and, per layer, a matrix  $W_{\text{struct}}$  whose output is added to the pre-activation of  $\Delta$  (Fig. 3):

$$\Delta_t = \text{softplus}(W_{\Delta}u_t + W_{\text{struct}}s_t + b_{\Delta}), \quad s_t = E(\varphi_t). \quad (4)$$

Two design choices make the comparison interpretable.  $W_{\text{struct}}$  is *zero-initialised*, so at step 0 the conditioned model computes exactly the baseline function and any divergence is learned. And the two arms



**Figure 3:** Structure enters at the timescale. The structural pathway (dashed) maps eight Hi-C features to a two-dimensional code and adds a learned bias to  $\Delta_t$  inside the softplus, so it modulates *retention* rather than content. Because  $W_{\text{struct}}$  is zero-initialised, the model is numerically identical to the sequence-only baseline at step 0.

are *paired*: a reference baseline is constructed under the same seed and copied into every shape-matching tensor, giving 275/275 bitwise-identical parameters at initialisation, so seed variance is shared rather than confounded with arm.

**Dual-scale conditioning.** §4 showed that a per-position input carries only  $\sim 11\%$  of  $\phi$ ’s variance, while a window-pooled input carries the complementary share but discards local detail. ChromaTau takes both. We form

$$\tilde{\phi}_t = \left[ \phi_t \parallel \frac{1}{|w|} \sum_{u \in w} \phi_u \right] \in \mathbb{R}^{16}, \quad (5)$$

concatenating the local feature vector with its mean over the enclosing window, and validity is the conjunction of both halves. The encoder input widens from 8 to 16 channels; nothing else in the architecture changes, because both the encoder and  $W_{\text{struct}}$  are pointwise and a pooled input is therefore equivalent to computing one embedding and broadcasting it.

This is the configuration we propose. The per-position variant is a strict subset of it ( $\tilde{\phi}$  with the second block ablated), so dual-scale conditioning cannot express less than per-position conditioning can.

**Removing the second ceiling.** The permeability gate of §5 imposes  $\tau \leq 1/p \approx 55$  tokens on the conditioned arm while the baseline, having no such term, pays nothing. We therefore disable it. This makes  $\tau$  denote the same quantity in both arms, which is a precondition for the comparison to mean anything; the gate had in any case been measured never to engage.

**Parameter accounting.** Baseline 7,725,312; ChromaTau 7,758,386 — a +0.428% difference against a 5% budget.

**Table 2:** Measured memory and throughput, single device, fp32, batch 2, with gradient checkpointing. Single process without gradient synchronisation; these are therefore a lower bound on distributed step time.

| window (bp) | baseline s/step | conditioned s/step | peak memory (GiB) |
|-------------|-----------------|--------------------|-------------------|
| 32,768      | 3.02            | 3.43               | 3.81              |
| 65,536      | 6.28            | 7.17               | 7.51              |
| 131,072     | 15.45           | 16.98              | 14.90             |

## 7 Experimental setup

**Data.** Human chromosome 9, GM12878, Hi-C from a published deeply-sequenced experiment set, mapped to GRCh38. Of 27,679 bins at 5 kb, 21,519 (77.7%) survive coverage filtering. Windows of 32,768 bp; 5,422 train, 274 validation, 243 test.

**Protocol.** Three seeds per arm, 2,000 steps, fp32 throughout, matched on steps and tokens rather than wall clock. Across-seed standard deviation of the baseline is  $\sigma = 0.0025$  bits, and the effect-size floor was fixed *in advance* at  $2\sigma = 0.0050$  bits.

**Controls.** At evaluation only, on the trained conditioned model, we substitute  $\phi$  with: S0 zeros; S1 a global permutation across bins; S2 a 10 Mb circular shift; S3 features recomputed from contacts resampled under a distance-matched null; S4 eight aligned one-dimensional sequence covariates. Controls draw from a fixed seed distinct from the training seed, so every training seed sees the same substituted structure.

**Computational cost.** Table 2 reports measured memory and step time, which bound what window widths are reachable and therefore what fraction of the structural signal is addressable at all (§4). Conditioning costs 10–14% in step time, from the two extra encoder passes required by the reverse-complement sign flip.

## 8 Results

**Preliminary study, and what it could not decide.** We first report a pilot conducted before the two design corrections of §5 and §6 were in place: per-position conditioning only, the permeability gate active, three seeds per arm, splits internal to one chromosome. Table 3 shows language-modelling loss unchanged, the conditioned arm 0.0020 bits worse at  $0.80\sigma$ .

**This pilot could not have produced a positive result, and we state that rather than interpret it.** Three factors independently prevent it: the conditioned arm carried the 55-token permeability ceiling the baseline does not, so the mechanism ran with  $\sim 8.5\times$  shorter memory than its comparator; conditioning was per-position, reaching  $\sim 11\%$  of the signal by construction (§4); and at three seeds per arm the exact permutation floor is 0.100, so  $p < 0.05$  was unreachable whatever the data showed. We report it because it motivated each correction, not as an evaluation of ChromaTau.

**Main evaluation.** The evaluation of ChromaTau as specified in §6 — dual-scale conditioning, permeability disabled, five paired seeds per arm, trained to a convergence criterion on a cross-chromosome split (train chr10–13, validate chr14–15, chr9 held out) — is in progress at submission time and its numbers are not reported here. The protocol, primary endpoint and falsification criterion were registered in advance and are stated in §7; at five paired seeds the permutation floor is  $2/252 = 0.008$ .

**Table 3:** Preliminary study, superseded. Per-position conditioning, permeability gate active, three seeds, within-chromosome splits. See the text for why this configuration could not express a positive result.

|                           | val bits/nt                           | median $\tau$ (as reported)           | per-seed                 |
|---------------------------|---------------------------------------|---------------------------------------|--------------------------|
| baseline                  | <b><math>1.5197 \pm 0.0025</math></b> | 434.7                                 | 1.5196 / 1.5172 / 1.5223 |
| per-position conditioning | $1.5217 \pm 0.0045$                   | 487.3                                 | 1.5266 / 1.5177 / 1.5209 |
| difference                | $+0.0020$ ( $0.80\sigma$ )            | permutation $p = 0.600$ (floor 0.100) |                          |

**Table 4:** Input-substitution controls at evaluation. Masking floor  $1.321 \times 10^{-1}$  is context for scale, not a threshold.

| control             | $\Delta$ bits | symmetric KL           | flip rate |
|---------------------|---------------|------------------------|-----------|
| s0 zeros            | $+0.0000$     | $2.539 \times 10^{-5}$ | 0.0052    |
| s1 permuted         | $+0.0001$     | $6.334 \times 10^{-5}$ | 0.0073    |
| s2 shifted          | $+0.0001$     | $5.528 \times 10^{-5}$ | 0.0068    |
| s3 distance-matched | $+0.0001$     | $6.827 \times 10^{-5}$ | 0.0075    |
| s4 sequence-matched | $+0.0000$     | $2.874 \times 10^{-5}$ | 0.0052    |

**The channel is connected but carries almost nothing.** Table 4 shows the input-substitution controls. Removing or corrupting  $\phi$  changes the output measurably — the structural input does reach the prediction — but the induced divergence is  $\sim 0.05\%$  of the divergence caused by masking itself. The ordering  $s3 \approx s1 \approx s2 \gg s4 \approx s0$  is what one expects if divergence simply tracks the distance of substituted values from the true ones, s0 being the standardised mean. It shows the input arrives; it does not show that structure is used.

**The pathway is trained but inert.** Gradient pressure reached the structural pathway: the Frobenius norm of  $W_{\text{struct}}$  is non-zero in 32/32 layer-directions in every seed (median 0.19–0.22) against the exact zero it was initialised to. Yet its contribution to the variance of  $\Delta$  is negligible. Defining  $D_1 = \text{Var}_t(W_{\text{struct}}s_t) / \text{Var}_t(W_{\Delta}u_t)$  with a pre-registered inertness threshold of 0.05, we measure pooled medians of 0.0004, 0.0002, 0.0002 across seeds.

The permeability gate never engaged: initialised at  $p = 0.018150$ , its measured mean over  $2.87 \times 10^8$  evaluations was 0.017234 / 0.017193 / 0.017108, with full range [0.0133, 0.0197] — all mass in a single bin.

**One probe does separate the arms.** Withholding  $\phi$  at probe time and predicting within-window insulation from hidden states after removing window identity, every conditioned seed exceeds every baseline seed (conditioned  $+0.0254$  /  $+0.0303$  /  $+0.0255$ ; baseline  $+0.0134$  /  $+0.0146$  /  $+0.0221$ ), attaining the 3v3 permutation floor of  $p = 0.100$ . Both arms sit *below* a local GC-composition floor of  $+0.0802$ , so this is relative rather than absolute structural sensitivity.

**External benchmarks.** To place the pretrained encoder against the literature we evaluate on a standard suite of genomic classification tasks [3] under a *frozen-encoder linear probe*: the encoder is fixed, sequences are mean-pooled, and a regularised logistic regression is fitted on top. Published numbers on these tasks are obtained by *fine-tuning* and are therefore not comparable; a frozen probe is a lower bound on the same weights. Since these sequences have no associated Hi-C — and some are not human — the conditioned arm is evaluated with  $\phi$  set to zero, which is the transfer setting that matters in practice.

On the pilot checkpoints, both arms exceed a  $k$ -mer composition floor ( $k \leq 3$ ) on every task, by  $+0.019$  and  $+0.033$  on the two with large test sets: pretraining on a single chromosome yields representations



beyond trinucleotide composition. The arms do not separate (mean difference  $+0.0019$  accuracy, no task beyond 1.6 standard errors), which is expected given the three limiting factors above. We note for methodological hygiene that the largest apparent gap falls on the smallest test set, where the binomial standard error is 0.0266 and the observed difference is 0.52 of it; across eight such tasks one arm sweeping the other by chance has probability 0.57, so we do not interpret it.

## 9 Discussion and limitations

**Scope.** We claim a mechanism, a measurement that motivates its design, and two initialisation results that make it expressible. We do not yet claim an improvement: the evaluation of ChromaTau as specified is in progress, and the preliminary study of §8 was run in a configuration that, as shown there, could not have produced a positive result.

**Limitations.** The scale is small: 7.7M parameters, one cell line,  $\approx 0.5$ B pretraining tokens. For calibration, a comparable published bidirectional-Mamba genomic model is parameter-matched to ours (7.73M,  $d_{\text{model}} = 256$ , 16 layers) but pretrained on roughly  $100\times$  more tokens; we compare mechanism, not scale. Structure comes from a single assay in a single cell type. And even dual-scale conditioning inherits the ceiling of §4 at the *inter*-window scale: structure varying across windows the model never sees together is out of reach for any architecture with bounded context.

**A caution for cross-cell-line evaluation.** The natural next test is generalisation to unseen cell types, and a common design probes insulation on a held-out line. We advise against it: TAD boundaries are substantially conserved, so such a probe can be passed by memorising the pretraining line. The usual remedy — probe A/B compartments instead, as more cell-type-variable — is conditional on the pair chosen. Measuring all eight features on a common chromosome across three lines, we find compartment agreement of  $+0.8389$  between GM12878 and K562 against insulation agreement of  $+0.7530$ : for this pair the premise inverts. It holds for GM12878 against IMR90 ( $+0.6906$  versus  $+0.7675$ ). The distinction tracks lineage. We therefore recommend targeting the *differential* between lines, on which a model that memorised the pretraining line scores zero by construction, and excluding contact-density features, which show the lowest agreement across *every* pair — the signature of depth and copy-number differences rather than regulatory architecture.

## 10 Conclusion

ChromaTau conditions the retention timescale of a state-space genomic model on measured chromatin structure during pretraining, rather than attaching structure to already-frozen representations. Its design follows from a measurement: roughly 90% of the structural signal’s variance lies between windows rather than within them — a property of Hi-C spatial autocorrelation that survives every change of window width and bin resolution available on current hardware — so conditioning must operate at two scales jointly. Making the mechanism expressible required removing a memory ceiling fixed at initialisation and present throughout this model family: the reference  $\Delta$  initialisation caps retention at 1,000 tokens before the first gradient step, and the fix costs no parameters while increasing trained retention thirty-fold. A second such constant can hide from the standard  $\tau$  diagnostic entirely, so that a model reports a memory horizon that is not its own — a failure mode we expect wherever selective state-space models are adapted to new domains.

## References

- [1] J. R. Dixon, S. Selvaraj, F. Yue, A. Kim, Y. Li, Y. Shen, M. Hu, J. S. Liu, and B. Ren. Topological domains in mammalian genomes identified by analysis of chromatin interactions. *Nature*, 485(7398):376–380, 2012.
- [2] A. Gu and T. Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv:2312.00752*, 2023.
- [3] K. Gresova, V. Martinek, D. Cechak, P. Simecek, and P. Alexiou. Genomic benchmarks: a collection of datasets for genomic sequence classification. *BMC Genomic Data*, 24(1):25, 2023.
- [4] E. Lieberman-Aiden, N. L. van Berkum, L. Williams, et al. Comprehensive mapping of long-range interactions reveals folding principles of the human genome. *Science*, 326(5950):289–293, 2009.
- [5] E. Nguyen, M. Poli, M. Faizi, et al. HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. *NeurIPS*, 2023.
- [6] S. S. P. Rao, M. H. Huntley, N. C. Durand, et al. A 3D map of the human genome at kilobase resolution reveals principles of chromatin looping. *Cell*, 159(7):1665–1680, 2014.
- [7] Y. Schiff, C.-H. Kao, A. Gokaslan, T. Dao, A. Gu, and V. Kuleshov. Caduceus: Bi-directional equivariant long-range DNA sequence modeling. *ICML*, 2024.
- [8] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. *NeurIPS*, 2017.